Pre-Algebraic Foundations for the CT Framework: Smoothly Undulating Numbers, Complexity Flow, and Algebraic Emergence

ODS Unified — Paper P00

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Abstract

We develop a pre-algebraic foundation for the Clifford–Thomas (CT) computational framework in which no algebraic structure is assumed *a priori*. The primitive objects are bounded oscillating series (smoothly undulating numbers). We introduce a complexity flow $\partial_\lambda u = G_\lambda[u]$ that preserves global bounds, admits local complexity increase, and defines limit attractors. We then define reading modes as integral operators and prove that, when the continuum of readings collapses to finitely many rigid classes, the limiting operators satisfy the Clifford anticommutation relations $\gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2g_{\mu\nu}\,\mathrm{Id}$. Thus Clifford algebra emerges as a boundary state of a purely analytic theory, rather than being postulated. All definitions are supported by 19 unit tests in the companion code.

1 Introduction

The CT runtime [1, 2, 3] currently operates on fields valued in the Clifford algebra $\mathrm{Cl}(3,0)$, an 8-dimensional graded algebra with generators satisfying $\gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2\delta_{\mu\nu}$. This algebraic structure is taken as axiomatic in P01–P03.

The present paper asks: *can the Clifford relations be derived rather than assumed?*

We answer affirmatively by constructing a purely analytic framework in which:

1. The only primitive objects are bounded oscillating series (no vectors, no matrices, no fixed dimension).
2. A complexity flow generates richness without breaking bounds.
3. Reading modes extract finite observations from the continuum.
4. When readings stabilize at discrete classes, Clifford structure emerges.

No algebra is used until it appears as a consequence.

2 Smoothly Undulating Numbers

Let $I = \mathbb{T}$ be a compact reading parameter. Fix a Schauder basis $(\phi_n)_{n\geq 0} \subset C^1(I)$. Define

$$\mathcal{U}_M := \left\{ u = \sum_{n\geq 0} c_n\phi_n \in C^1(I) : \|u\|_\infty \leq M \right\}.$$

Definition 1 (Analytic oscillator)**.** *A function* $\omega \in C^2(\mathbb{R})$ *is an* analytic oscillator *if:* $\sup|\omega| = 1$, $\inf \omega = -1$, $\omega(\xi + P) = \omega(\xi)$ *for some* $P > 0$, $\int_0^P \omega(\xi)\, d\xi = 0$, *and all critical points of* ω *are isolated. No trigonometry is required.*

Definition 2 (Smoothly undulating number)**.** *An element* $u \in \mathcal{U}_M$ *is a* smoothly undulating number *(SUN) if there exists a factorization*

$$u(s) = \alpha(s)\, \omega(\varphi(s)),$$

with $\alpha \in W^{1,\infty}(I)$, $0 \le \alpha \le M$*;* $\varphi \in C^1(I)$, $\varphi'(s) \ge \kappa > 0$*;* ω *an analytic oscillator; and the slowness condition*

$$|\alpha'(s)| \le \vartheta\,(1 + \alpha(s))\,\varphi'(s), \qquad 0 \le \vartheta < 1. \tag{A1}$$

The triple $(\alpha, \omega, \varphi)$ *is an* admissible factorization, *not the object itself.*

Definition 3 (Equivalence of factorizations)**.** *Two factorizations* $(\alpha, \omega, \varphi)$ *and* $(\tilde\alpha, \tilde\omega, \tilde\varphi)$ *represent the same SUN if there exists* $h \in C^1(\mathbb{R})$, *strictly increasing, with* $h' > 0$, *such that* $\tilde\varphi = h \circ \varphi$, $\tilde\omega = \omega \circ h^{-1}$, $\tilde\alpha = \alpha$.

Proposition 1 (Phase reparametrization invariance)**.** *The relation of Definition 3 is an equivalence relation and leaves* u *invariant.*

Proof. $\tilde\alpha(s)\,\tilde\omega(\tilde\varphi(s)) = \alpha(s)\,\omega(h^{-1}(h(\varphi(s)))) = \alpha(s)\,\omega(\varphi(s)) = u(s)$. Reflexivity, symmetry, and transitivity are immediate. □ □

Proposition 2 (Series compatibility)**.** *Every smoothly undulating number belongs to the permitted ontological class: it is a bounded series.*

Proof. By definition $u \in C^1(I) \subset C(I)$. Since (ϕ_n) is a Schauder basis of $C(I)$, there exists a unique expansion $u = \sum_{n \ge 0} c_n \phi_n$ with uniform convergence. The bound $\|u\|_\infty \le M$ holds by construction. □ □

Proposition 3 (Sequences as readings)**.** *For any* $(b_n)_{n \ge 0} \in \ell^\infty$, *there exist* $u \in \mathcal{U}_M$ *with* $M = \sup_n |b_n|$ *and smooth readers* (L_n) *such that* $L_n[u] = b_n$ *for all* n.

Proof. Take disjoint intervals $J_n \subset I$ and smooth functions ψ_n, θ_n supported in J_n with $\int_I \psi_n \theta_n = 1$ and $\int_I \psi_m \theta_n = 0$ for $m \ne n$. Set $u = \sum_n b_n \psi_n$ and $L_n[v] = \int_I v\, \theta_n$. Then $\|u\|_\infty \le M$ by disjoint support, and $L_n[u] = b_n$. □ □

3 Complexity Flow

Fix two smoothing operators (heat semigroup):

$$S_c := e^{\varepsilon_c \partial_{ss}}, \quad S_f := e^{\varepsilon_f \partial_{ss}}, \quad 0 < \varepsilon_f < \varepsilon_c.$$

Define the *defect* $A[u] := u - S_c u$, the *fine return* $\mathcal{R}[u] := S_f u - S_c u$, the *complexity density* $Q[u] := |A[u]|$, and the *production proxy* $\Pi[u] := |\mathcal{R}[u]|$.

Definition 4 (Dynamic permeability)**.** *For bounded continuous* $\alpha, \beta, \mu : [0, \infty) \to (0, \infty)$ *and* $\tau > 0$*:*

$$p_\lambda[u] := \beta(\lambda)\, \frac{\Pi[u]}{\Pi[u] + \mu(\lambda)\, Q[u] + \tau}. \tag{B1}$$

Then $0 \le p_\lambda[u] \le \beta(\lambda)$, *and the local rupture threshold is* $\Pi[u] > \mu(\lambda)\, Q[u] + \tau$.

Definition 5 (Complexity flow)**.**

$$\partial_\lambda u = G_\lambda[u] := -\alpha(\lambda)\, A[u] + p_\lambda[u] \left(1 - \frac{u^2}{M^2}\right) \mathcal{R}[u]. \tag{F}$$

Proposition 4 (Global existence and boundedness). *For any $u_0 \in \mathcal{U}_M$, the flow* (F) *has a unique global solution in $C^1([0,\infty); C(I))$ satisfying $\|u(\lambda)\|_\infty \le M$ for all $\lambda \ge 0$.*

Proof. G_λ is locally Lipschitz in $C(I)$ because the denominator in (B1) is bounded below by $\tau > 0$. Existence follows from Picard–Lindelöf in Banach space.

For the bound: if $u(\lambda_*, s_*) = M$ at a first contact point, then $A[u] = u - S_c u \ge 0$ (since S_c is positive and sub-averaging), so the first term in (F) is ≤ 0. The second term vanishes because $1 - u^2/M^2 = 0$. Thus $\partial_\lambda u \le 0$ at the boundary. The argument for $-M$ is symmetric. Therefore $[-M, M]$ is an invariant tube, preventing blow-up. □ □

Proposition 5 (Complexity production identity). *Define the complexity energy $\mathcal{E}[u] := \frac{1}{2}\int_I u\,\mathcal{R}[u]\,ds$. Along the flow* (F)*:*

$$\frac{d}{d\lambda}\mathcal{E}[u(\lambda)] = -\alpha(\lambda)\int_I A[u]\,\mathcal{R}[u]\,ds + \int_I p_\lambda[u]\left(1 - \frac{u^2}{M^2}\right)\Pi[u]^2\,ds. \tag{B3}$$

Moreover, $\int_I A[u]\,\mathcal{R}[u]\,ds \ge 0$.

Proof. Since $\mathcal{R}$ is self-adjoint, $\frac{d}{d\lambda}\mathcal{E} = \int_I \partial_\lambda u\,\mathcal{R}[u]\,ds$. Substituting (F) and using $|\mathcal{R}[u]| = \Pi[u]$ yields (B3). For the positivity: $A = I - S_c$ and $\mathcal{R} = S_f - S_c$ are increasing functions of the same heat operator, hence commute and are both positive, giving $\int A\,\mathcal{R} \ge 0$. □ □

Corollary 1 (Rupture and complexity increase). *Define the rupture set $\Omega_{\mathrm{rup}}(\lambda) := \{s : \Pi[u] > \mu(\lambda)Q[u] + \tau\}$. On Ω_{rup}, the permeability satisfies $p_\lambda[u] > \beta(\lambda)/2$. If $|u| < M$ on a positive-measure subset of Ω_{rup} and local production exceeds dissipation, then $\frac{d}{d\lambda}\mathcal{E}[u(\lambda)] > 0$.*

Definition 6 (Limit attractor). *Assuming $\alpha(\lambda) \to \alpha_\infty$, $\beta(\lambda) \to \beta_\infty$, $\mu(\lambda) \to \mu_\infty$ as $\lambda \to \infty$, a* limit attractor *is a stable solution $u_* \in \mathcal{U}_M$ of the stationary equation $G_\infty[u_*] = 0$.*

Conjecture 1 (Generic convergence). *For an open set of smoothly undulating initial data and regular schedules (α, β, μ), the flow converges to a non-empty family of limit attractors with finite rigid rank.*

4 Algebraic Emergence

Definition 7 (Reading modes). *For a compact parameter set Ξ with $\xi = (\eta, \sigma, \delta)$ and resolution $\varepsilon > 0$, a* reading mode *is an integral operator*

$$(L_\xi^\varepsilon u)(s) = \int_I K_\xi^\varepsilon(s, r)\,u(r)\,dr,$$

with smooth kernel K_ξ^ε satisfying uniform boundedness. These are not *ontological layers; they are readers with focus η, scale σ, and penumbra δ.*

Definition 8 (Quantization radius).

$$\Delta_N(\varepsilon; u) := \inf_{v_1,\dots,v_N \in C(I)} \sup_{\xi\in\Xi} \min_{1\le j\le N} \|L_\xi^\varepsilon u - v_j\|_\infty.$$

Definition 9 (Rigid rank). $N_{\mathrm{rig}}(u) := \min\{N \ge 1 : \lim_{\varepsilon\to 0} \Delta_N(\varepsilon; u) = 0\}$, *if it exists. A finite rigid rank means the continuum of readers collapses to N discrete classes.*

Definition 10 (Symmetrized closure defect). *For class representatives $L_1^\varepsilon, \dots, L_N^\varepsilon$ of the N rigid classes, define*

$$D_{\mu\nu}^\varepsilon[u; c] := \sup_{\substack{v\in\mathcal{H}_\varepsilon[u] \\ \|v\|_\infty\le 1}} \left\| L_\mu^\varepsilon L_\nu^\varepsilon v + L_\nu^\varepsilon L_\mu^\varepsilon v - 2c\,v \right\|_\infty,$$

where $\mathcal{H}_\varepsilon[u]$ is the family generated by iterated reading. Let $g_{\mu\nu}^\varepsilon[u]$ be any minimizer in c, and $D_{\mu\nu}^\varepsilon[u] := D_{\mu\nu}^\varepsilon[u; g_{\mu\nu}^\varepsilon]$.

Definition 11 (Rigidity functional)**.**

$$\mathbf{Rig}_N(\varepsilon; u) := \Delta_N(\varepsilon; u) + \max_{1\le\mu,\nu\le N} D^{\varepsilon}_{\mu\nu}[u].$$

First term: discretization. Second term: algebraic closure.

Definition 12 (Algebraic emergence of rank N)**.** *u exhibits* algebraic emergence of rank N *if* $N_{\mathrm{rig}}(u) = N$ *and* $\lim_{\varepsilon\to 0} \mathbf{Rig}_N(\varepsilon; u) = 0$.

Theorem 1 (Recovery of Clifford relations)**.** *Let u_* be a limit attractor of the flow. Suppose:*

1. $N_{\mathrm{rig}}(u_*) = N < \infty$.
2. $\mathbf{Rig}_N(\varepsilon; u_*) \to 0$.
3. *For each μ, the strong limit $\gamma_\mu v := \lim_{\varepsilon\to 0} L^{\varepsilon}_{\mu} v$ exists for all $v \in \mathcal{H}[u_*]$.*
4. *The limit $g_{\mu\nu} := \lim_{\varepsilon\to 0} g^{\varepsilon}_{\mu\nu}[u_*]$ exists.*

Then

$$\gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2g_{\mu\nu}\,\mathrm{Id} \quad \text{on } \mathcal{H}[u_*]. \tag{1}$$

Proof. By Definition 10, $\|L^{\varepsilon}_{\mu}L^{\varepsilon}_{\nu}v + L^{\varepsilon}_{\nu}L^{\varepsilon}_{\mu}v - 2g^{\varepsilon}_{\mu\nu}v\|_\infty \to 0$ for all $v \in \mathcal{H}_\varepsilon[u_*]$ with $\|v\|_\infty \le 1$. Passing to the strong limits in the operators and the scalar limit in $g^{\varepsilon}_{\mu\nu}$ yields (1). □ □

Corollary 2 (Squares ± 1)**.** *If $g_{\mu\mu} \neq 0$, the renormalized reader $\widetilde{\gamma}_\mu := |g_{\mu\mu}|^{-1/2}\gamma_\mu$ satisfies $\widetilde{\gamma}_\mu^2 = \mathrm{sgn}(g_{\mu\mu})\,\mathrm{Id}$. If $g_{\mu\nu} = 0$ for $\mu \neq \nu$, then $\widetilde{\gamma}_\mu\widetilde{\gamma}_\nu + \widetilde{\gamma}_\nu\widetilde{\gamma}_\mu = 2\eta_{\mu\nu}\,\mathrm{Id}$ with $\eta = \mathrm{diag}(\pm 1, \ldots, \pm 1)$.*

Corollary 3 (Cl(3,0) sub-closure)**.** *If $N_{\mathrm{rig}}(u_*) = 8$ and there exist three classes $\{\mu_1, \mu_2, \mu_3\}$ with $g_{\mu_i\mu_j} = \delta_{ij}$, then an emergent sub-closure of type $\mathrm{Cl}(3,0)$ exists within the attractor's eight reading modes. This matches the current ODS runtime without assuming eight coordinates* a priori.

5 Experiments

All definitions and propositions are accompanied by computational tests in the companion code `_claude_calculo/`.

Test group	**Tests**	**Status**
Def 1 (oscillator)	3	3/3
Def 2 (SUN)	2	2/2
Prop 1 (invariance)	1	1/1
Prop 3 (sequences)	1	1/1
Prop 4 (boundedness)	3	3/3
Prop 5 (energy)	2	2/2
Cor 1 (rupture)	2	2/2
Emergence (Defs 9–14)	5	5/5
Total	**19**	**19/19**

Table 1: Test coverage for the pre-algebraic foundations.

6 Discussion

6.1 What this paper does

- Provides a purely analytic foundation for CT.
- Derives Clifford structure as a consequence, not an axiom.
- Defines exact criteria for the continuum-to-discrete transition.
- Preserves global bounds throughout the complexity flow.

6.2 What this paper does not claim

- Generic convergence of the flow to rigid attractors (Conjecture 1, not proven).
- That the emergent metric $g_{\mu\nu}$ necessarily matches any physical metric.
- That this framework resolves any open problem in PDE theory.
- That the specific reading mode kernels used in the experiments are the "correct" or unique choice.

7 Conclusion

We have shown that Clifford algebra can emerge from a purely analytic framework of bounded oscillating series, without postulating any algebraic structure. The key mechanism is the collapse of a continuum of reading modes to finitely many rigid classes under resolution refinement. When this rigidity condition is met (Definition 12), the limiting operators satisfy the Clifford anticommutation relations (Theorem 1).

This provides a foundational justification for the CT runtime series P01–P06: the $\mathrm{Cl}(3,0)$ structure used there is not arbitrary but arises as a natural boundary state of the complexity flow.

The single open problem is Conjecture 1: proving generic convergence of the flow to finite-rank rigid attractors.

References

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